

waetr: An R Package for Web Accessibility Evaluation at Scale

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Editor: 

Submitted: 05 May 2026

Published: unpublished

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Summary

waetr is an R package that automates large-scale evaluation of website accessibility. It connects to the WAVE Web Accessibility Evaluation Tools API, allowing users to batch-submit URLs, retrieve counts of accessibility features and errors, and organize the results into tidy data frames for analysis. waetr simplifies authentication, handles request formatting and response parsing, and provides functions for summarizing and visualizing accessibility patterns across multiple web pages.

Statement of Need

Web accessibility is essential for individuals with visual, hearing, motor, and/or cognitive disabilities to access and benefit from online content (Yesilada et al., 2015). Approximately 15% of the world's population lives with some form of disability, making web accessibility a critical concern for ensuring equitable access to digital resources (World Health Organization, 2011). Inaccessible web pages create barriers for individuals with disabilities across education, healthcare, employment, and civic participation (W3C, 2018). The Web Content Accessibility Guidelines (WCAG) define testable criteria for web pages such as color contrast, keyboard operability, font size, and alternative image text (W3C, 2023). Adherence to the WCAG improves web usability and navigability across the general population, not just individuals with disabilities (Petrie & Kheir, 2007).

Previous approaches to web accessibility evaluation have relied on manual testing, automated tools with limited scalability, or custom scripts that lack reproducibility. While researchers have developed methods for large-scale accessibility evaluation (Ara & Sik-Lanyi, 2025), these often involve non-public code that cannot be readily adopted by the research community. The WebAIM Million project demonstrates the value of large-scale accessibility analysis (WebAIM, 2023a), but the tools used for such analyses have not been made available in a format accessible to HCI researchers.

WebAIM's WAVE API evaluates web pages based on WCAG testable criteria (WebAIM, 2023b), providing counts of accessible and inaccessible page elements. Currently, the WAVE API and browser extension support sequential evaluation of web page accessibility (i.e., one web page at a time). Although this is useful for developers and organizations seeking to improve web content accessibility during product development, there are several significant limitations for researchers seeking to evaluate web accessibility more broadly. Given the growing adoption of R in HCI research, as evidenced by specialized textbooks (Robertson & Kaptein, 2016), statistical packages (Wobbrock, 2024), and educational resources (Yatani, n.d.), the waetr package addresses these limitations by providing a comprehensive, open-source solution for large-scale accessibility evaluation.

41 Additionally, web accessibility is not just a moral obligation; it is also a legal requirement
42 in many countries ([U.S. Department of Justice, 2022](#)). Laws such as the Americans with
43 Disabilities Act (ADA) in the United States and the European Accessibility Act in the European
44 Union mandate that websites must be accessible to individuals with disabilities. The waetr
45 package helps organizations evaluate compliance with these regulations more efficiently.

46 Features

47 The waetr R package provides solutions to current limitations and challenges through the
48 following features:

- 49 1. **Batch Processing:** The WAVE Chrome extension and web interface currently support
50 sequential evaluation of individual webpages, limiting the ability to efficiently evaluate
51 larger volumes of web pages ([WebAIM, 2023c](#)). waetr solves this limitation via batch
52 processing, facilitating the evaluation of multiple web pages concurrently.
- 53 2. **Data Export:** The WAVE web interface and browser extension output lack effective
54 means for exporting accessibility evaluation information in formats suitable for further
55 analysis and reporting. This creates barriers for researchers needing to process and
56 interpret accessibility data at scale. The waetr package addresses this limitation
57 by providing comprehensive data export capabilities, generating HTML reports with
58 interactive visualizations, CSV summary data, PNG plot files, and detailed accessibility
59 metrics for saving, sharing, and analyzing accessibility findings through preferred data
60 analysis tools and workflows.
- 61 3. **Data Structure:** The WAVE web interface provides graphical depictions of
62 compliance/non-compliance features through visual icons and counts, but offers no
63 exportable data. Although the API can translate this visual information into numerical
64 data, the raw output is not appropriately structured for statistical analyses. The waetr
65 package transforms this data into tidy, analysis-ready formats.
- 66 4. **Simplified Integration:** Numerous steps are required to extract and prepare WAVE data
67 to ensure compatibility with software to conduct statistical analyses or data visualization.
68 The waetr package eliminates this complexity by providing direct functions for data
69 retrieval, processing, and visualization within the R environment.
- 70 5. **Research Translation:** The gap that prevents translation from individual site evaluation
71 to large-scale analyses hinders advancement in the field of web accessibility research.
72 waetr bridges this gap by enabling researchers to conduct comprehensive, data-driven
73 accessibility studies across multiple websites, making it possible to identify patterns and
74 trends in web accessibility at scale.

75 Conclusion

76 The waetr package is a valuable tool for researchers studying web accessibility and developers
77 committed to creating accessible and inclusive websites. By providing an easy-to-use interface
78 to the WebAIM WAVE API, the package enables efficient evaluation and analysis of website
79 accessibility at scale.

80 In a world where web accessibility is increasingly important, the waetr package empowers both
81 researchers to conduct meaningful accessibility studies and developers to create websites that
82 are usable by all individuals. By facilitating web accessibility evaluation through tools like
83 waetr, we can work towards building a more inclusive and equitable online landscape.

84 Recent research has shown that automated accessibility evaluation tools, when properly
85 integrated into research workflows, can significantly advance our understanding of web
86 accessibility patterns and trends ([Brajnik et al., 2012](#); [Campoverde-Molina et al., 2021](#)).

- 87 The waetr package contributes to this ongoing effort by providing a reproducible, open-source
88 solution that can be adopted and extended by the research community. By contributing to the
89 open-source community and sharing knowledge and resources, we can collectively make the
90 web a more accessible place for everyone.
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